2.2.2 Determine the ideal shaping time

You will use a single shaping time throughout this experiment. This section will help you select the ideal shaping time at the beginning.

- Adjust the coarse gain and fine gain such that the alpha peak occurs around the center of the spectrum.
- For each shaping time option on your amplifier, perform the following steps:
  - Record a ~1 minute run in Maestro
  - Set an ROI around the peak (drag your mouse over the alpha peak and select mark ROI). Be sure that the ROI includes the entire peak, but do not extend it too far to either side.
  - Double click the ROI and record the centroid channel and gross area of the alpha peak.

Now pick the best shaping time based on what your learned in this course to date. You will want to select the shortest shaping time that does not exhibit a significant ballistic deficit. Confirm with GSI for the shaping time you chose.

2.2.3 Determine a suitable gain and LLD for the counting curves

The gain and LLD will depend on the range of pulses produced in each section. Betas produce a broad spectrum and require a higher gain, while alphas produce a more peaked spectrum and require lower gain.

First, we will set the gain and LLD for the Alpha source, and then these steps will be repeated for the Beta source.

In this lab, we will use the MCA to generate the counting curve. This means that instead of using an LLD setting we will choose a channel number as the LLD and only include counts above this channel number.

GAIN:

- You should be using the shaping time determined in the previous section. Adjust the gain such that the alpha source peak at 800V is near to the maximum MCA channel.

LLD:

In general, we want the LLD set such that all counts are discriminated at the minimum voltage setting. This means the LLD must be set above the maximum of the noise and any source counts generated. For this measurement, it is sufficient to start at ~300V.

- Set the voltage to 300V.
- Start collecting data and note the maximum channel including either source or noise counts.
- Choose a channel number as the LLD such that all counts are discriminated.

2.3 Counting Curve Measurements

NOTE: The system is very sensitive to changes in high voltage and you should allow the system to stabilize (~ 30 s) after changing the voltage before measuring the count rate. Ensure that your counting curve begins below the lowest voltage that generates visible pulses from the output of the amplifier, which can be observed on the scope. Make sure the high voltage is varied over a range such that all the features of the counting curve can be observed. Collect enough data points to distinguish the features of the counting curve. Set the counting time be long enough that uncertainty in counts is low. **DO NOT EXCEED 1700 V**.

2.3.1 Generation of the counting curve using an alpha source

- First, we will determine the count rate vs. voltage characteristic of the counter using a source that only emits alphas, such as ^{241}Am.